Problem 8

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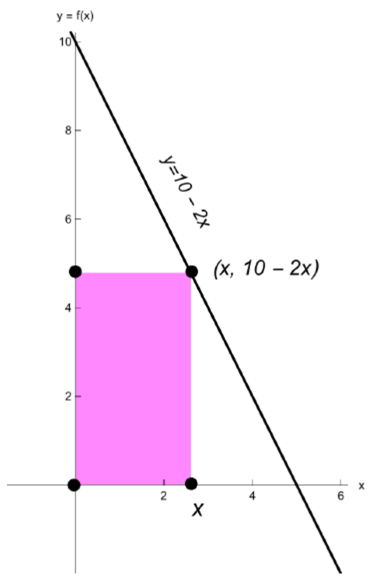
Problem 1 *A rectangle is constructed with one side on the positive x-axis, one side on the positive y-axis, and the vertex opposite the origin on the line $y = 10 - 2x$. What dimensions maximize the area of the rectangle? What is the maximum area?*

Explanation. The four vertices of the rectangle are $(0, 0)$, $(x, 0)$, (x, y), and $(0, y)$:



We're trying to maximize the area $A = x \cdot y$.

First, we want to turn this area formula into a function of x. To do that we use the constraint $y = 10 - 2x$. Substituting, we find the function we're trying to maximize is $A(x) = x(10 - 2x) = 10x - 2x^2$.

Next, we need to identify the domain of A.

To find the largest value we compute the x-intercept of $y = 10 - 2x$:

$$10 - 2x = 0 \implies x = 5.$$

Therefore $\text{Domain}(A) = (0, 5)$.

Next we locate the critical points of A. Since $A'(x) = 10 - 4x$ is defined on $(0, 5)$, to locate critical points solve

$$\begin{aligned} A'(x) = 0 &\iff 10 - 4x = 0 \\ &\iff x = \frac{5}{2}. \end{aligned}$$

To finish, we check the sign of the derivative an intervals $(0, \frac{5}{2})$, and $(\frac{5}{2}, 5)$. Since $A'(x) = 10 - 4x = -4(x - \frac{5}{2})$, it follows that $A'(x) > 0$ on the interval $(0, \frac{5}{2})$ and $A'(x) < 0$ on the interval $(\frac{5}{2}, 5)$. Therfore the function A has a local maximum at $x = \frac{5}{2}$. This also a global maximum, since the function is increasing on $(0, \frac{5}{2})$, and decreasing on $(\frac{5}{2}, 5)$. Therefore the maximum area occurs when the rectangle has a base with length $5/2$ units and height with length of 5 units. The area with these dimensions is $25/2$.